SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0509**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	**Question**	**Bloom's Level**
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

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Code of Conduct:

I SANDRA SUDEVAN_192424422 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Sandra Sudevan
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema

A schema is the overall design or structure of a database. It defines how data is organized.

Example:

Student(RollNo, Name, Department, Age)

This structure remains the same unless modified.

Instance

An instance is the actual data stored in the database at a particular time.

ROLL NO	NAME	DEPARTMENT	AGE
101	Anu	CSE	20
102	Ragul	IT	19

The data may change, but the schema usually remains the same.

Difference between Schema and Instance

Schema	Instance
Structure of database	Actual data in database
Changes rarely	Changes frequently
Defined by DBA	Updated by users
Created during database design	Changes during database operations

Types of Schema

1. Logical Schema

- Describes tables, attributes, and relationships.
- Independent of storage.

Example:

Student(RollNo, Name, Age)

2. Physical Schema

- Describes how data is stored in memory.
- Includes indexes, file organization, storage blocks.

3. View Schema

- Shows only required data to a user.
- Provides security.

Example:

Student(Name, Department)

Advantages of Schema

- Provides a clear database structure.
- Reduces data redundancy.
- Ensures data consistency.
- Supports data integrity.
- Simplifies database maintenance.
- Improves security through views.

Advantages of Instance

- Reflects real-time database information.
- Changes dynamically according to user operations.
- Enables data retrieval and updates.
- Supports day-to-day database transactions.

Conclusion

A schema defines the structure of a database, while an instance represents the actual data stored at a particular moment. The logical schema defines the database design, the physical schema describes how data is stored, and the view schema provides customized access to users, ensuring security and ease of use.

2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Entities and Attributes

Customer

- Customer_ID (PK)
- Name
- Address
- Phone

Account

- Account_No (PK)
- Account_Type
- Balance

Branch

- Branch_ID (PK)
- Branch_Name
- Location

Loan

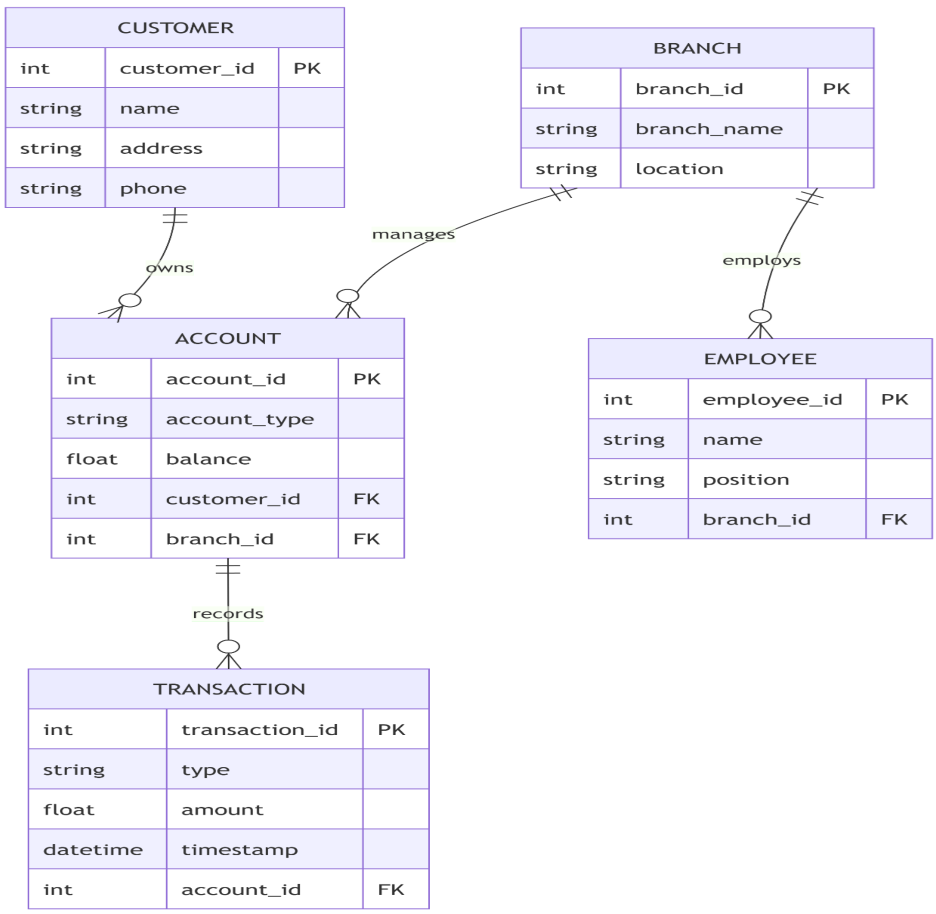
- Loan_ID (PK)

- Amount
- Loan_Type

Relationships

- Customer **owns** Account
- Branch **maintains** Account
- Customer **takes** Loan
- Branch **provides** Loan

ER Diagram (Simple)



3. Explain the Extended ER (EER) model. Describe generalization, specialization, and aggregation.

Extended ER (EER) Model

The EER model is an extension of the ER model. It supports advanced concepts like inheritance and grouping.

Features

- Generalization
- Specialization
- Aggregation
- Inheritance

Generalization

Generalization combines similar lower-level entities into a higher-level entity.

Example

Student

↓

Teacher

↓

⬇ Person

Both Student and Teacher become Person.

Specialization

Specialization divides one entity into smaller entities.

Example

Employee

⬇

⬇ ⬇

Manager Engineer

Employee is divided into Manager and Engineer.

Aggregation

Aggregation treats a relationship as a higher-level entity.

Example

Employee ---- Works_On ---- Project

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Aggregation

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Department

Aggregation is useful when relationships participate in another relationship.

4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Entities

Patient

- Patient_ID (PK)
- Name
- Age

Doctor

- Doctor_ID (PK)
- Name
- Specialization

Appointment

- Appointment_ID (PK)
- Date
- Time

Bill (Weak Entity)

- Bill_No
- Amount

Inheritance

```
    Person
       |
|               |
Patient         Doctor
```

Person is the superclass.

Patient and Doctor are subclasses.

Weak Entity

```
Patient
  |
Has
  |
Bill
```

Bill depends on Patient for its existence.

Simple EER Diagram

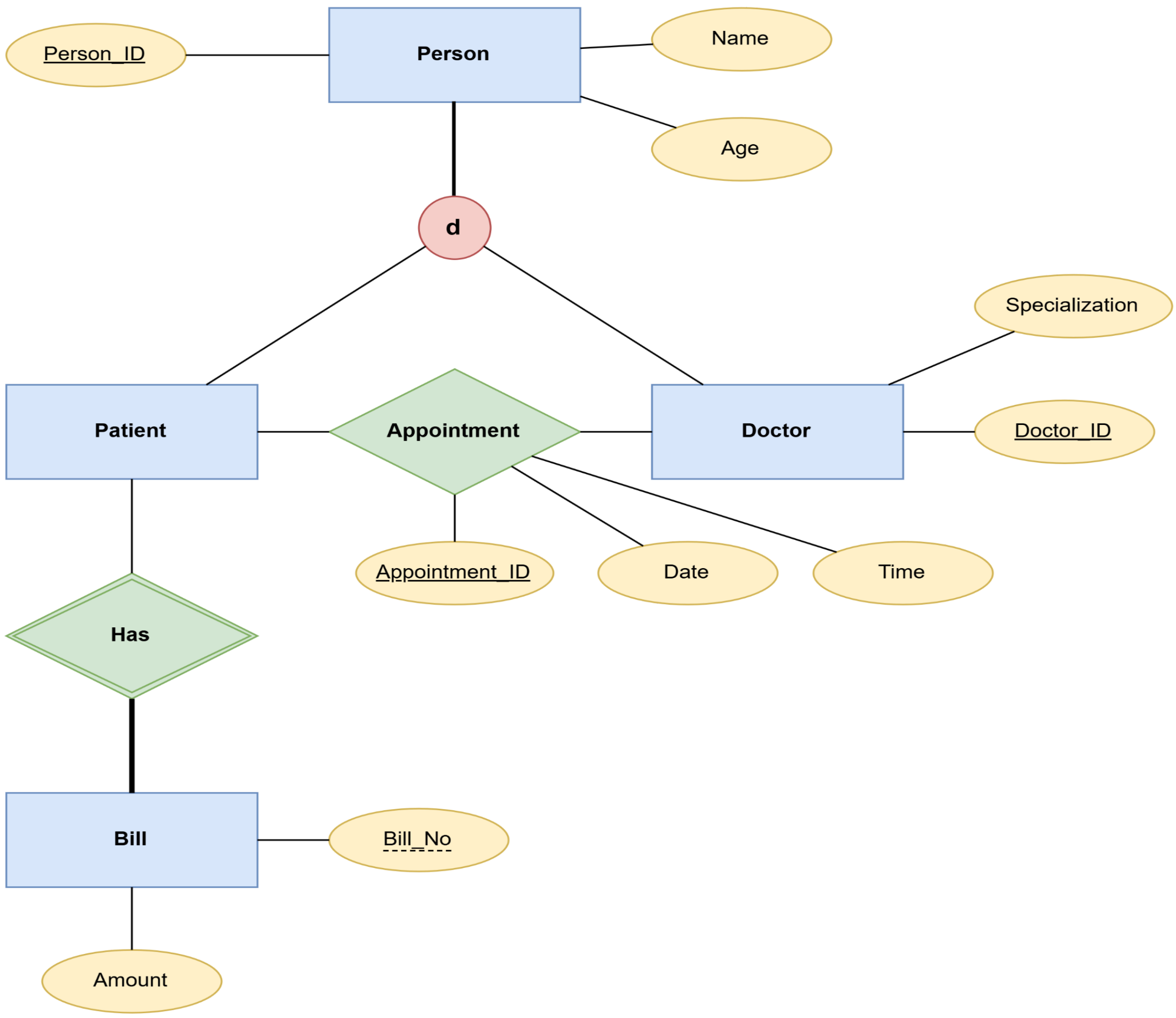
```
        Person
        /       \
        Patient         Doctor


Patient -------- Appointment -------- Doctor
  |
```

Has

|

Bill (Weak Entity)



5. Explain the role of DBA (Database Administrator) and the responsibilities associated with the position.

Database Administrator (DBA)

A Database Administrator (DBA) is responsible for managing, securing, and maintaining a database system.

Responsibilities

1. Install and configure DBMS.
2. Create and manage databases.
3. Manage users and permissions.
4. Ensure database security.
5. Perform backup and recovery.
6. Monitor database performance.

7. Optimize queries and storage.
8. Maintain data integrity.
9. Handle database failures.
10. Upgrade and maintain the DBMS.

Advantages of DBA

- Ensures data security.
- Improves database performance.
- Prevents data loss.
- Maintains database availability.
- Supports multiple users efficiently.

Conclusion

A DBA ensures that the database remains secure, reliable, efficient, and available for all authorized users.